

Short-Term Memory

→ Refers to the information that an AI agent keeps only for current interaction or few past recent conversations.

→ Temporary & very limited in size (just like RAM)

Characteristics of Short-Term Memory

→ Temporary

→ Contextual

↳ focuses on current task/conversation (or)
active messages

→ limited

→ fast access.

→ Session oriented

→ Personalized

Long-Term Memory

→ Refers to the information that an AI agent stores for a longer duration often across session & interaction.

→ User preferences
Personal details
Pattern
Knowledge.

Characteristics of Long-Term Memory

→ Permanent

→ Durable / Persistent

→ Selective

↳ Not everything should be stored. Only meaningful, sensible & relevant information should be stored.

→ Personalized

Short-Term (vs) Long-Term Memory

Aspects	Short-Term	Long-Term
Duration	temporary	persistent
scope	current session	across sessions.
Size	Small	large
Purpose	immediate context	reusable information
Lifespan	Often ends with session	can last <u>forever</u> .
Usage	Current discussion	<u>Personalization.</u>

Conversational Memory

Memory that agent remembers & use this during a conversation so that the interaction feels more connected & natural.

→ Mostly associated to short-term.

Memory Storage Strategies.

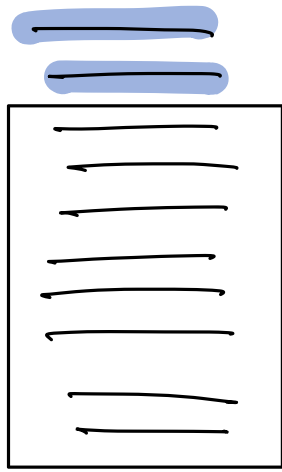
Strategy - 1 : Store recent conversation history.

→ Store few recent messages

→ Short-term.

→ Simple & effective for maintaining current Context.

→ Old information may get lost when the conversation becomes long.



Strategy - 2: Summarize past Conversation

→ Compact storage

→ Some info might be lost in summarization.

Strategy - 3: Key: Value Pairs.

→ Structured way to manage memory

→ Personalized & preferences info.

Working Memory

- Actively used portion of the memory used for current moment.
- associated with short term memory.

Episodic Memory

- Stores data about recent events, interactions or experiences.
- associated with long-term memory.

Semantic Memory

- general knowledge, fact & figures.
- associated with long-term memory.

Procedural Memory

- knowledge about how to do things.
- workflows, access patterns, rules or procedures.
- associated with long-term memory.

Short-Term Memory Strategies.

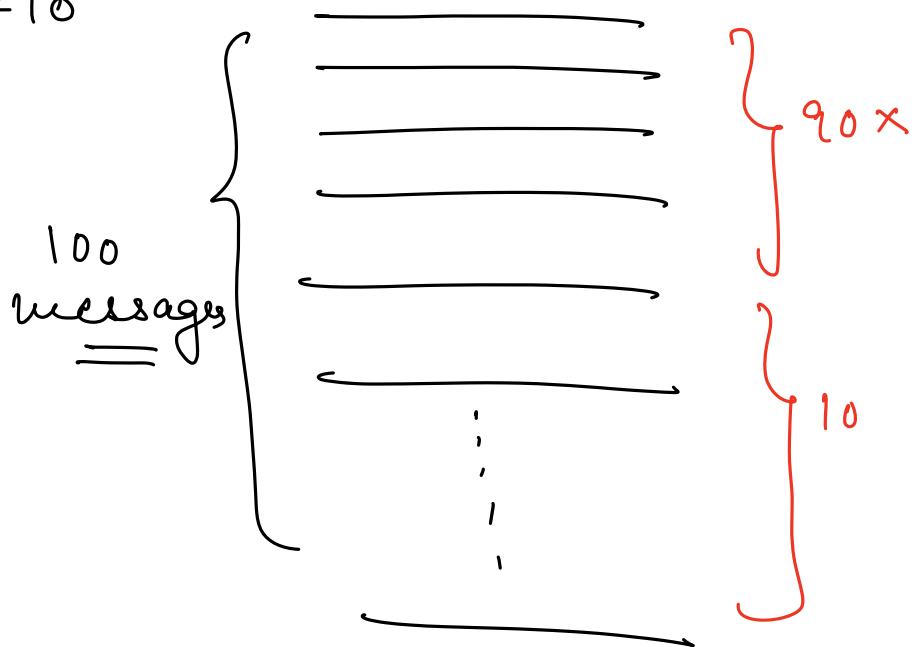
1. Buffer memory Strategy

- Agent stores the full recent conversations / messages as is.
- Best for short conversations.
- Memory can grow very fast & it starts losing the context from the old conversations.

2. Window memory Strategy

- Agent only stores last N messages.

window size = 10



Summary memory Strategy

- Agent doesn't store all the conversations / messages, it only stores the summary of past. Conversations.
- Space efficient
- Good for long conversations.
- Some details may get missed.